Course Name: Computer Networks and Internet Protocol

Assignment - Week 11 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following addresses is used in Ethernet broadcast for ARP requests?

a) 255.255.255.255
b) 255.255.255.254
c) FF:FF:FF:FF:FF:FF
d) 255.255.255.251

Correct Answer: (c)
Detailed Solution: When a host sends an ARP request, it does not know the destination MAC address, so it sends the frame to all devices in the LAN using the broadcast MAC address: FF:FF:FF:FF:FF:FF.

QUESTION 2

Which field in an ARP packet is not filled in ARP Request but filled in ARP Reply?

a) Target protocol address
b) Target hardware address
c) Sender protocol address
d) Sender hardware address

Correct Answer: (b)
Detailed Solution: In an ARP request, the sender already knows its own IP and MAC address and also the target IP address, but it does not know the target MAC address. Therefore, the target hardware address field is left empty in the request and is filled in the ARP reply.

QUESTION 3

Which protocol does DHCP use at the Transport layer?

a) IP
b) ARP
c) UDP
d) TCP

Correct Answer: (c)
Detailed Solution: DHCP operates at the application layer but uses the UDP protocol at the transport layer for communication (ports 67 for server and 68 for client), since it requires connectionless and fast message exchange.

QUESTION 4

Which of the following is/are correct about RARP?

a) RARP replies are broadcast.
b) RARP is often used on thin-client workstations.
c) In an RARP Request packet, the sender hardware address field is not filled.
d) RARP has been replaced by DHCP.

Correct Answer: (b, d)
Detailed Solution: RARP replies are unicast, not broadcast, and the sender hardware address is included in the request. It was used by thin clients and has been replaced by DHCP.

QUESTION 5

To deal with hidden terminal and exposed station problems, all implementations of 802.11 WLAN must mandatorily support _____________.

a) DCF
b) PCF
c) Both are mandatory
d) Both are optional

Correct Answer: (a)
Detailed Solution: In IEEE 802.11 WLAN, DCF (Distributed Coordination Function) is mandatory and uses mechanisms like CSMA/CA to handle hidden and exposed terminal problems. PCF is optional and not required in all implementations.

QUESTION 6

When a new trunk link is configured on an IOS-based switch, which VLANs are allowed over the link?

a) By default, all VLANs are allowed on the trunk.
b) No VLAN's are allowed, you must configure each VLAN by hand.
c) Only configured VLAN's are allowed on the link.
d) Only extended VLAN's are allowed by default.

Correct Answer : (a)
Detailed Solution: On an IOS-based switch, when a trunk link is configured, all VLANs are allowed by default unless explicitly restricted by configuration.

QUESTION 7

Which VLAN technique is used when a single link needs to carry traffic for multiple VLANs?

a) Tailing
b) Trunking
c) Trading
d) Tapping

Correct Answer: (b)
Detailed Solution: When a single link needs to carry traffic for multiple VLANs, trunking is used, which allows multiple VLAN frames to pass over the same physical link using tagging.

QUESTION 8

A switch has been configured for three different VLANs: VLAN2, VLAN3, and VLAN4. A router has been added to provide communication between the VLANs. What type of interface is necessary on the router if only one connection is to be made between the router and the switch?

a) 10Mbps Ethernet
b) 56Kbps Serial
c) 100Mbps Ethernet
d) 1Gbps Ethernet

Correct Answer: (c)
Detailed Solution: For inter-VLAN communication using a single link between the switch and router (router-on-a-stick), the link must support trunking and sufficient bandwidth. A 100 Mbps Ethernet interface is the minimum suitable option.

QUESTION 9

Which of the following specifies a set of media access control (MAC) and physical layer specifications for implementing WLANs?

a) IEEE 802.16
b) IEEE 802.3
c) IEEE 802.11
d) IEEE 802.15

Correct Answer: (c)
Detailed Solution: IEEE 802.11 is a set of media access control and physical layer specification for implementing WLAN computer communication. It was founded in 1987 to begin standardization of spread spectrum WLANs for use in the ISM bands

---

QUESTION 10

In a LAN of 100 hosts, each host sends an ARP request once every 10 seconds. Each ARP packet is 64 bytes (including Ethernet frame). What is the total ARP traffic generated per second?

a) 320 bytes/sec
b) 640 bytes/sec
c) 1280 bytes/sec
d) 6400 bytes/sec

Correct Answer: (b)
Detailed Solution: Packet per second = (100/10) = 10 packet/sec. Total traffic = (10x64) = 640 bytes/sec.
